3. Model Development and Evaluation Report

3.1 Dataset Splitting

- **Train-Test Split Ratio:**
  - **80%** training dataset
  - **20%** testing dataset
- **Outputs:**
  - x_train, x_test

  - y_train, y_test

3.2 Models Implemented

1. **Logistic Regression**
   - Baseline linear classification model.
   - Highly interpretable and provides clear baseline coefficient weights.
2. **Decision Tree Classifier**
   - Non-parametric tree-based classifier.
   - Effectively captures non-linear relationships and interactions without rigid distribution assumptions.
3. **Random Forest Classifier**
   - Ensemble bagging classifier built on multiple decision trees.
   - Reduces variance and guards against overfitting.
4. **Gaussian Naive Bayes**
   - Probabilistic classifier based on Bayes' theorem.
   - Fast, effective with standardized features, and strong performance on imbalanced recall tasks.
5. **K-Nearest Neighbors (KNN)**
   - Instance-based non-parametric classifier using distance metrics.
   - Predicts class based on feature-space proximity to k nearest neighbors.

3.3 Model Training & Validation

- **Cross-Validation Framework:**
  - **5-Fold Stratified Cross-Validation** (StratifiedKFold(n_splits=5, shuffle=True)) to maintain class balance across training folds.
- **Standard Training Protocol:**
  - Feature matrices scaled via StandardScaler.
  - Models evaluated on scaled training partitions using cross_validate().

- Final chosen model trained on full scaled training set using best_model.fit(x_train, y_train).

3.4 Model Evaluation

Metrics Used (Classification)

- **Accuracy:** Overall proportion of correctly predicted instances.
- **Precision:** Ratio of true positives to total predicted positives ($\frac{TP}{TP + FP}$).
- **Recall:** Ratio of true positives to total actual positive cases ($\frac{TP}{TP + FN}$).
- **F1-Score:** Harmonic mean of Precision and Recall ($\frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$), serving as the primary metric due to target class imbalance.
- **Confusion Matrix & Classification Report:** Detailed breakdown of true positives, false positives, true negatives, and false negatives.

3.5 Results Summary (5-Fold Cross-Validation Metrics)

Model Name	Accuracy	Precision	Recall	F1-Score
Gaussian Naive Bayes	**77.6%**	**40.1%**	**63.1%**	**48.9%**
Logistic Regression	86.4%	67.6%	37.3%	48.0%
Decision Tree	78.7%	37.1%	37.9%	37.4%
Random Forest	85.2%	71.5%	20.2%	31.4%
K-Nearest Neighbors	84.1%	62.9%	14.7%	23.7%

3.6 Model Selection

- **Selected Model: Gaussian Naive Bayes** (best_model = models['Naive Bayes']).

Reason for Selection

- **Highest F1-Score:** Achieving the top average **F1-score (~0.489)** across 5-fold cross-validation among all candidate models.

- **Superior Recall Performance:** Achieved a **63.1% Recall** (compared to Random Forest's 20.2% and Logistic Regression's 37.3%), ensuring the majority of actual attrition cases (flight risks) are successfully identified.
- **Business Alignment:** In HR employee attrition modeling, missing an employee who is about to leave (False Negative) carries a much higher business cost than flagging a retained employee for engagement (False Positive). Naive Bayes maximizes detection of actual at-risk personnel.